

Quadratic Stability without Subtraction: Unique-Base Semirings, Scalar Extension, and Thermograph Walls

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Abstract

Over a semiring, polarization, cancellation, and group completion are distinct operations. We separate them by working with finite free quadratic pairs (q, b) over semirings with unique bases. The pair is encoded by a weighted coefficient graph, monomial basis change acts by a unit gauge, and orthogonal sum is disjoint union. Consequently the isometry monoid is free commutative on connected gauge-orbit types. Regular pairs split into rank-one loops and rank-two edges, and basic metabolic stabilization deletes exactly the metabolic edge generators.

For the closed Hessenberg fragment $\mathcal{H}_d = \{\alpha < \omega^{\omega^d}\}$, natural sum and product identify $\mathcal{H}_d$ with $\mathbb{N}[x_0, \dots, x_{d-1}]$. Its only unit is one and additive cancellation makes the companion unique and balanced. Every regular pair is consequently a sum of copies of one hyperbolic plane, so the regular Witt group is zero. The all-pairs stable monoid nevertheless remains free on every connected weighted-graph type except the hyperbolic edge and has an exact finite decision procedure.

For the standard rational tangible supertropical cover we give the analogous gauge-orbit presentation, the complete rank-two companion ambiguity, and the regular metabolic generators. Coefficientwise supertropicalization is functorial only for based forms. For every field of characteristic different from two and every rational tangible supervaluation, the universal basis-independent ring-image quotient is exactly $\mathbb{Z}/2$, detected by rank parity, for both the canonical balanced and polar-companion lifts.

Finally, the same framework tests a proposed game-valued application. Completing thermograph walls by one global tropical zero produces honest max-plus and min-plus coefficient semifields, and each wall pair defines a regular rank-one quadratic pair classified by its gap. The self-negative hot switch $\{1 \mid -1\}$ has nonzero gap, so this stable wall class cannot be additive under disjunctive sum.

1 Introduction

Classical Witt theory uses subtraction at several logically different points: to recover a polar form from a quadratic form, to cancel orthogonal summands, and to pass from isometry classes to a group. None of these moves is automatic over a semiring. The correct first object is therefore a quadratic pair (q, b) , where

$$q(ax) = a^2q(x), \quad q(x + y) = q(x) + q(y) + b(x, y),$$

and b is a specified symmetric bilinear companion [6]. Isometries in this paper preserve both entries of the pair.

The organizing input is the unique-base theorem of Izhakian, Knebusch, and Rowen: if the nonzero elements of a semiring are closed under addition and multiplication, every basis of a free module differs from every other by a permutation and multiplication by scalar units [5, Theorem 1.3]. Matrix reduction is thereby replaced by finite graph theory. Connected components

control orthogonal decomposition, while invertible Gram matrices reduce to involutions of the unique basis. This one mechanism governs both the regular theory and its much larger singular envelope.

We use that mechanism to answer one question in three coefficient worlds: what does stable quadratic structure retain when subtraction is unavailable? Hessenberg natural arithmetic gives a rigid, cancellative example; supertropical arithmetic adds nonunique companions and a meaningful scalar-extension problem; thermograph walls test whether the resulting stable classes can carry addition from the Conway game group.

The main conclusions are as follows.

Theorem A. *Let stable equivalence kill finite free regular pairs carrying a basic Lagrangian, as in Definition 2.4.*

1. *For every $d \geq 0$, all regular quadratic pairs over the Hessenberg fragment $\mathcal{H}_d$ are hyperbolic. Its regular Witt group is zero. The stable monoid of all pairs is the free commutative monoid on connected weighted-graph types other than the hyperbolic plane.*
2. *Over the standard rational tangible supertropical cover, the all-pairs stable group is free abelian on connected coefficient-graph gauge orbits other than the regular metabolic edge orbits. Regular pairs have only loop and edge blocks, and rank-two companion ambiguity is explicit.*
3. *Canonical coefficientwise supertropicalization neither descends from based forms to unbased isometry classes nor preserves regularity under an arbitrary classical change of basis. For a source field of characteristic different from two, imposing all classical basis relations makes its universal nonsingular ring-image quotient exactly $\mathbb{Z}/2$, detected by rank parity. The same is true for the alternative polar-companion lift.*
4. *The max/min thermograph walls do define regular rank-one stable classes over completed piecewise-linear tropical semifields. Their class is the wall gap, but it cannot be additive in the Conway game group.*

The proof has a single spine. Section 2 constructs the graph presentation and stable quotient for an arbitrary unique-base semiring. Sections 3 and 4 specialize it to Hessenberg and rational supertropical coefficients. Section 5 applies the same rank-one classification to thermograph walls. Section 6 states exactly which joins are kernel-checked and which remain cited or paper-level mathematics.

The Hessenberg result is a complete classification and collapse theorem. The supertropical block theorem and universal classical ring-image quotient are complete for the stated rational tangible cover. The wall result is a no-go theorem for the natural max/min construction; it does not rule out every possible synchronized quotient that deliberately kills the exhibited gap class.

2 Quadratic pairs over unique-base semirings

Let R be a commutative semiring such that $R \setminus \{0\}$ is closed under addition and multiplication. Let $V = R^n$ with basis $E = (e_1, \dots, e_n)$, and let $P = (V, q, b)$ be a quadratic pair.

2.1 Coefficient graphs and cancellation

Definition 2.1 (Coefficient graph). The coefficient graph $G_E(P)$ has vertex set $\{1, \dots, n\}$. Vertex i is labelled by

$$(a_i, d_i) = (q(e_i), b(e_i, e_i)).$$

For $i < j$, there is an edge ij precisely when $c_{ij} = b(e_i, e_j) \neq 0$, and that edge is labelled by c_{ij} . Only labelings that occur in a quadratic pair are called admissible.

If $u_i \in R^\times$ and $\pi \in S_n$, the monomial basis change $e_i \mapsto u_i e_{\pi(i)}$ acts by

$$a_i \mapsto u_i^2 a_{\pi(i)}, \quad d_i \mapsto u_i^2 d_{\pi(i)}, \quad c_{ij} \mapsto u_i u_j c_{\pi(i)\pi(j)}.$$

We call this the unit gauge action.

Proposition 2.2 (Coordinate expansion). *For $x = \sum_i x_i e_i$,*

$$q(x) = \sum_i x_i^2 a_i + \sum_{i < j} x_i x_j c_{ij}.$$

Thus $G_E(P)$, including the diagonal labels d_i , determines the pair.

Proof. Apply the second quadratic-pair identity repeatedly. Bilinearity gives $b(x_i e_i, x_j e_j) = x_i x_j c_{ij}$, and homogeneity gives $q(x_i e_i) = x_i^2 a_i$. A bilinear form is determined by its matrix on the basis, so the additional labels d_i determine the full companion. $\square$

Theorem 2.3 (Graph presentation). *Let $\mathcal{C}_R$ be the set of connected admissible coefficient graphs modulo the unit gauge action and labelled-graph isomorphism. The monoid of isometry classes of finite free quadratic pairs satisfies*

$$\mathcal{P}(R) \cong \mathbb{N}^{(\mathcal{C}_R)}.$$

In particular, orthogonal cancellation holds.

Proof. The unique-base theorem makes every module automorphism monomial. By Proposition 2.2, such a map is an isometry of pairs exactly when its permutation and units obey the displayed gauge law. An orthogonal decomposition along subsets of the basis is equivalent to the absence of cross edges. Hence the indecomposable orthogonal summands are exactly the connected components of the coefficient graph. Finite labelled graphs have unique connected-component multisets, which gives the displayed free commutative monoid and cancellation. This is the pair-valued specialization of the component and cancellation theory in [5, Theorems 2.6 and 4.9]. $\square$

Write $b^\flat : V \rightarrow V^* = \text{Hom}_R(V, R)$ for $b^\flat(x) = b(x, -)$.

2.2 Regular blocks and stabilization

Definition 2.4 (Regular, split, and metabolic). A pair is *regular* if $b^\flat$ is an isomorphism. A submodule is *basic* if it is spanned by a subset of the unique basis. A regular pair is *basic metabolic* if it has a basic submodule L such that $q|_L = 0$ and $L = L^\perp$. It is *basic split* if it has complementary basic Lagrangians.

Let $\mathcal{N}_R$ be the monoid of basic metabolic pairs. Define

$$A \sim_{\text{st}} B \iff \exists N_1, N_2 \in \mathcal{N}_R : A \perp N_1 \cong B \perp N_2.$$

The all-pairs stable group is

$$W_{\text{all}}(R) = K_0(\mathcal{P}(R)) / \langle \mathcal{N}_R \rangle.$$

The regular Witt group is defined by the same construction inside the regular submonoid.

Allowing different neutral objects on the two sides is essential. Adding the same object to both sides gives no new equivalence because Theorem 2.3 already proves cancellation.

Theorem 2.5 (Regular block theorem). *Every regular quadratic pair over R is an orthogonal sum of blocks of two kinds:*

$$\begin{aligned} A(a, d) : \quad & q(e) = a, \quad b(e, e) = d \in R^\times, \\ H(a_1, a_2; c) : \quad & b = \begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}, \quad c \in R^\times. \end{aligned}$$

A regular pair is basic metabolic exactly when it has no loop blocks and each edge block has $a_1 = 0$ or $a_2 = 0$. It is split exactly when every block is $H(0, 0; c)$.

Proof. If $b^\flat$ is invertible, its matrix sends a basis of V to a basis of V^* . Unique bases make it a generalized permutation matrix. Symmetry makes the underlying permutation an involution. A fixed point gives a loop block; a transposition gives an edge block.

Let L be basic with $L = L^\perp$. A loop basis vector can neither lie in L , because it is not self-orthogonal, nor outside L , because it would then lie in $L^\perp \setminus L$. In an edge block, equality with the orthogonal complement forces L to contain exactly one endpoint. The condition $q|_L = 0$ says that the selected endpoint has zero quadratic label. Complementary Lagrangians exist exactly when both endpoint labels vanish. $\square$

Consequently, whenever admissibility is decidable, the stable invariant is decidable: decompose the graph, canonically compare the finitely many gauge orbits, delete the metabolic connected edge types, and sort the remaining multiset.

3 Hessenberg natural arithmetic

Let $\#$ and $\otimes$ denote Hessenberg natural sum and product. Their Cantor-normal-form definitions add coefficients and formally multiply monomials, using natural sum on the exponents [3].

3.1 Polynomial coordinates and companion rigidity

Definition 3.1 (Closed finite-CNF fragment). For $d \in \mathbb{N}$, put

$$\mathcal{H}_d = \{\alpha : \alpha < \omega^{\omega^d}\}.$$

It is equipped with $\#$ and $\otimes$, not ordinary ordinal addition or multiplication and not Conway nim arithmetic.

Proposition 3.2 (Polynomial model). *There is a semiring isomorphism*

$$(\mathcal{H}_d, \#, \otimes) \cong \mathbb{N}[x_0, \dots, x_{d-1}].$$

Hence addition is cancellative, nonzero elements are closed under both operations, $\mathcal{H}_d^\times = \{1\}$, $a\#a = 1$ has no solution, and $a\#a = 0$ implies $a = 0$.

Proof. Every exponent $\beta < \omega^d$ has the unique expansion $\beta = \sum_{k < d} \omega^k n_k$. Send ω^β to $\prod_{k < d} x_k^{n_k}$, and extend by the finite outer Cantor normal form. Natural sum adds like coefficients, while natural product adds exponent vectors and multiplies coefficients. This proves the isomorphism. All listed properties are immediate in a polynomial semiring over $\mathbb{N}$. $\square$

Fix a quadratic pair over $\mathcal{H}_d$. We continue to write addition and multiplication in semiring notation.

Lemma 3.3 (Unique balanced companion). *The companion is uniquely determined by q , and*

$$b(x, x) = 2q(x)$$

for every x .

Proof. If b and c are companions, cancel $q(x) + q(y)$ from their two instances of the quadratic-pair identity. For balance, compute $q(x + x)$ in two ways:

$$4q(x) = 2q(x) + b(x, x).$$

Additive cancellation gives $b(x, x) = 2q(x)$. □

Thus a rank-two pair has the unique form

$$Q(a, c, d)(x, y) = ax^2 + cxy + dy^2, \quad b = \begin{pmatrix} 2a & c \\ c & 2d \end{pmatrix}.$$

3.2 Regular collapse and the singular envelope

Proposition 3.4 (Rank two). *The isometry class of $Q(a, c, d)$ is the orbit of (a, c, d) under $(a, c, d) \leftrightarrow (d, c, a)$. It is decomposable exactly when $c = 0$, and regular exactly when*

$$(a, c, d) = (0, 1, 0).$$

Proof. The only scalar unit is one, so every rank-two automorphism is either the identity or the coordinate swap. The graph is disconnected exactly when its one possible edge is absent. If the Gram matrix is invertible, unique bases make it a permutation matrix. Its diagonal entries $2a, 2d$ cannot equal one. Thus it must be the transposition matrix, which forces the displayed triple. The converse is direct. □

Write $H = Q(0, 1, 0)$.

Theorem 3.5 (Hessenberg regular collapse). *Every regular finite free quadratic pair over $\mathcal{H}_d$ is isometric to $H^{\perp m}$ for a unique m . Regular, split, and basic metabolic pairs coincide. Consequently*

$$\mathrm{GW}_{\mathrm{reg}}(\mathcal{H}_d) \cong \mathbb{Z}[H], \quad W_{\mathrm{reg}}(\mathcal{H}_d) = 0.$$

Proof. Theorem 2.5 reduces the issue to its permutation orbits. A loop would have $1 = b(e, e) = 2q(e)$, impossible by Proposition 3.2. Hence the permutation is fixed-point-free. On a transposition the only nonzero Gram entries are the two off-diagonal ones, both equal to the only unit, one. Its diagonal equality $0 = 2q(e)$ forces both quadratic labels to vanish. Every orbit is therefore H . The remaining assertions follow from the block theorem. □

The singular envelope does not collapse. Let $\mathcal{C}_d$ be the set of connected $\mathcal{H}_d$ -weighted graph types from Definition 2.1.

Corollary 3.6 (Complete stable invariant).

$$\mathcal{P}(\mathcal{H}_d)/\sim_{\mathrm{st}} \cong \mathbb{N}^{(\mathcal{C}_d \setminus \{H\})}, \quad W_{\mathrm{all}}(\mathcal{H}_d) \cong \mathbb{Z}^{(\mathcal{C}_d \setminus \{H\})}.$$

There are no further relations.

Proof. Theorem 2.3 gives the free monoid on all connected types, and Theorem 3.5 says that the metabolic submonoid is freely generated by the single type H . Quotienting and group-completing deletes exactly that generator. $\square$

For example, $Q(0, 2, 0)$ and H have identical diagonal data, but the former is a connected generator different from H and survives stably. Thus the singular invariant contains genuine off-diagonal information beyond the forced Cantor-normal-form labels. An exact decision procedure normalizes all CNFs, finds connected components, compares their finitely many vertex permutations, discards H , and compares the sorted multisets.

There is a useful warning about definitions. If regularity is omitted and “metabolic” means only that some basic L satisfies $q|_L = 0$ and $L = L^\perp$, then every $Q(0, c, 0)$ with $c \neq 0$ becomes metabolic: take $L = \mathcal{H}_d e_1$. When c is not a unit the polar map is not surjective. Such a definition would erase singular edge weights for an accidental semiring reason, so regularity or perfect cross-pairing is indispensable.

4 Rational supertropical quadratic pairs

4.1 Intrinsic blocks and companion ambiguity

Let $U_{\mathbb{Q}}$ be the standard tangible max-plus supertropical cover

$$\{0\} \sqcup \{t^\gamma : \gamma \in \mathbb{Q}\} \sqcup \{(t^\gamma)^\nu : \gamma \in \mathbb{Q}\}.$$

Unequal values add by dominance; a tie becomes the corresponding ghost. The tangible elements t^γ are exactly the scalar units. Write $e = 1 + 1$ for the ghost of one. The value group is dense and two-divisible. The min-plus statement is obtained by $\gamma \mapsto -\gamma$. Free modules have unique bases [6, Theorem 0.9].

For a pair, label vertex i by (a_i, d_i) and edge ij by c_{ij} , as in Definition 2.1. Theorem 2.3 gives the following complete presentation.

Theorem 4.1 (Supertropical stable blocks). *The pair monoid over $U_{\mathbb{Q}}$ is free commutative on connected admissible coefficient graphs modulo*

$$(a_i, d_i, c_{ij}) \mapsto (u_i^2 a_{\pi(i)}, u_i^2 d_{\pi(i)}, u_i u_j c_{\pi(i)\pi(j)}).$$

Every regular pair is an orthogonal sum of loop blocks $A(a, d)$ and edge blocks $H(a_1, a_2; c)$ from Theorem 2.5. The all-pairs stable group is free abelian on all connected types except the edge types

$$H(0, a; c), \quad H(a, 0; c), \quad c \in U_{\mathbb{Q}}^\times,$$

which are exactly the connected basic metabolic types.

This is a generators-and-relations theorem with no hidden relations. It also gives a decision procedure. For each of finitely many permutations, write $u_i = t^{s_i}$. Equality of the value labels becomes a finite rational linear system in the s_i , together with equality of zero patterns and of tangible/ghost layers. Canonical connected-orbit representatives can therefore be compared in every finite rank.

Companion ambiguity is already visible in rank two. Present a functional quadratic form as

$$q = \begin{bmatrix} a_1 & \alpha \\ & a_2 \end{bmatrix}.$$

Proposition 4.2 (Complete rank-two companions). *All companions have matrices*

$$b = \begin{pmatrix} d_1 & c \\ c & d_2 \end{pmatrix}, \quad d_i \in [0, ea_i],$$

where

$$c \in \begin{cases} \{\alpha\}, & a_1 a_2 = 0, \\ \{\alpha\}, & \alpha^2 >_\nu a_1 a_2, \\ \{z : z^2 \leq_\nu a_1 a_2\}, & a_1 a_2 \neq 0 \text{ and } \alpha^2 \leq_\nu a_1 a_2. \end{cases}$$

The choices of d_1, d_2, c are independent.

Proof. The diagonal intervals are [6, Proposition 6.12]. If one diagonal coefficient vanishes, the off-diagonal companion is rigid by Proposition 7.9 there. Otherwise the dense, square-divisible specialization of Theorems 7.11 and 7.12 gives the remaining two cases. Independence of the matrix entries is Theorem 6.3 of the same source. $\square$

In particular, one functional q can admit companions with different regular block structures. Stable theory over a supertropical semiring must therefore specify pairs, not only functional quadratic forms.

4.2 The scalar-extension obstruction

Let K be a field of characteristic different from two and let $\varphi : K \rightarrow U_{\mathbb{Q}}$ be a tangible supervaluation into the standard rational cover. Thus $\varphi(a) = t^{v(a)}$ for $a \neq 0$, where $v : K^\times \rightarrow \mathbb{Q}$, and the tangible unit group of $U_{\mathbb{Q}}$ is square-divisible. Coefficientwise supertropicalization is functorial for an ordered based form and monoidal under orthogonal sum [6, Definition 9.3]. It is not an invariant of the unbased form.

Consider the classical hyperbolic plane

$$q(xe + yf) = xy.$$

In the basis (e, f) , its supertropicalization with the canonical balanced companion is the split edge $H(0, 0; 1)$, hence has stable class zero. In the equally valid basis

$$v_1 = e + f, \quad v_2 = e - f,$$

one has $q(rv_1 + sv_2) = r^2 - s^2$. Equations (9.11)–(9.13) of [6] give

$$D(1, e) \perp D(1, e), \quad D(a, ea) : q(\epsilon) = a, \quad b(\epsilon, \epsilon) = ea.$$

The ghost e is not a scalar unit, so both loop blocks are nonregular. They are nevertheless connected generators in the all-pairs group and are not metabolic. Therefore

$$[H(0, 0; 1)] = 0, \quad 2[D(1, e)] \neq 0.$$

Corollary 4.3 (No unbased canonical extension). *Canonical coefficientwise supertropicalization does not descend to unbased isometry classes, even after regular basic-metabolic stabilization. Moreover, the regular category is not closed under changing the classical input basis.*

Remark 9.5 of [6] supplies a distinct choice: tropicalize the classical polar companion itself. Its diagonal is $\varphi(2a_i)$, rather than $e\varphi(a_i)$, and it is still a companion. In the changed basis above this alternative produces two regular nonmetabolic loops; it repairs regularity in this example but not basis independence. This distinction is also the basis-dependence phenomenon studied explicitly in [7, Section 8].

We now impose exactly the missing relations. For $\epsilon \in \{\text{bal}, \text{pol}\}$, let $S_\epsilon(K, \varphi)$ be the subgroup of $W_{\text{all}}(U_{\mathbb{Q}})$ generated by the based lifts $T_\epsilon(q, E)$ of nonsingular classical forms. Let $\overline{S}_\epsilon(K, \varphi)$ be its quotient by all relations

$$[T_\epsilon(q, E)] = [T_\epsilon(q, E')]$$

for two bases of the same form, together with classical hyperbolic stabilization. This is the universal target for an additive basis-independent invariant extracted from the indicated coefficientwise lift.

Theorem 4.4 (Universal ring-image collapse). *For every K, φ as above,*

$$\overline{S}_{\text{bal}}(K, \varphi) \cong \mathbb{Z}/2, \quad \overline{S}_{\text{pol}}(K, \varphi) \cong \mathbb{Z}/2.$$

Both isomorphisms are rank modulo two. Consequently no square class, discriminant, Hasse invariant, signature, Witt index, coefficient valuation, or residue datum survives the universal unbased ring-image quotient.

Proof. Diagonalize a nonsingular form:

$$q \cong \langle a_1, \dots, a_n \rangle, \quad a_i \neq 0.$$

For each i , choose a tangible unit u_i with $u_i^2 = \varphi(a_i)^{-1}$. In the canonical balanced lift, scaling the i -th basis vector by u_i normalizes its rank-one pair to

$$D : \quad q(\epsilon) = 1, \quad \beta(\epsilon, \epsilon) = e.$$

Hence every n -dimensional canonical image is basis-related to $D^{\perp n}$.

For the polar-companion lift, multiplicativity gives

$$\varphi(2a_i) = \varphi(2)\varphi(a_i).$$

The same scaling normalizes every line to the single pair

$$A_{\varphi(2)} : \quad q(\epsilon) = 1, \quad b(\epsilon, \epsilon) = \varphi(2).$$

Thus the polar image of every n -dimensional form is basis-related to $A_{\varphi(2)}^{\perp n}$. In particular, valuations and square classes change neither generator.

For the hyperbolic plane $q(xe + yf) = xy$, the basis (e, f) gives the split edge $H(0, 0; 1)$, which is zero in the stable group. The basis $(e + f, e - f)$ gives $r^2 - s^2$, hence two copies of D , respectively two copies of $A_{\varphi(2)}$. Therefore the relevant generator L obeys $2L = 0$, proving that each quotient is at most $\mathbb{Z}/2$.

Rank modulo two is additive, unchanged by basis change, and zero on every basic metabolic pair. It sends L to one, so $L \neq 0$ and no stronger relation is possible. This also proves the stated universal property. $\square$

The theorem concerns the subgroup generated by classical scalar-extension images. It does not collapse the intrinsic group $W_{\text{all}}(U_{\mathbb{Q}})$: connected noncanonical weight or layer types that no nonsingular classical lift realizes remain independent free generators. In particular, the isolated

zero line survives in the ambient group even though it is absent from the nonsingular ring-image subgroup.

There is also a sharp regularity distinction. The polar lift has a regular diagonal presentation, and its regular ring-image quotient is again $\mathbb{Z}/2$. A generic polar presentation is nevertheless nonregular. For the canonical balanced lift, a nonzero diagonal companion coefficient is ghost and not a unit. A regular canonical ring image must therefore have zero diagonal and perfect-matching off-diagonal support, which makes the classical form hyperbolic and the image split. Thus its regular ring-image subgroup is zero; the all-pairs category in Theorem 4.4 is essential.

If degenerate classical forms are included, the standard cover still leaves only total-rank parity. Indeed, the polar matrices

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

have identical coefficientwise images, while their ranks are one and two. After classical diagonalization this identifies the zero line Z with L ; hence every finite free form has class $\text{rank}(V)L$.

Proposition 4.5 (Trivially valued characteristic two). *Let K have characteristic two and let $\varphi(0) = 0$, $\varphi(K^\times) = 1$. For the canonical balanced and polar-companion lifts considered separately, the nonsingular universal ring-image quotient is*

$$\begin{cases} \mathbb{Z}/2, & K = \mathbb{F}_2, \\ 0, & |K| > 2. \end{cases}$$

The surviving $\mathbb{F}_2$ -class is the Arf invariant.

Proof. The polar form is nondegenerate alternating, so a symplectic basis decomposes q into planes

$$a_i x^2 + xy + b_i y^2.$$

Only the zero patterns of (a_i, b_i) remain after φ . The $(0, 0)$ -plane is split. A plane with exactly one nonzero endpoint is also killed: for the balanced lift this follows by comparing the bases (e, f) and $(e, e + f)$ of $q = xy$, while for the polar lift it is already basic metabolic.

It remains only the plane with both endpoints nonzero. If $|K| > 2$, choose $\lambda \notin \{0, 1\}$. The hyperbolic basis

$$e + f, \quad e + \lambda f$$

has two nonzero quadratic labels and nonzero mutual polar label, so the remaining generator is itself zero. Over $\mathbb{F}_2$, that generator is the anisotropic plane $x^2 + xy + y^2$. Its coefficient support determines the Arf bit [2]. For two copies with bases (e_1, f_1, e_2, f_2) , the vectors

$$f_1 + f_2, \quad f_1 + e_2, \quad e_1 + e_2 + f_2, \quad e_1 + f_1 + e_2 + f_2$$

are isotropic and form two mutually orthogonal symplectic pairs. Thus two anisotropic planes are hyperbolic, giving order two, while Arf proves that one does not vanish. Symplectic decomposition proves the assertion in every rank. $\square$

5 Thermograph walls and additivity

5.1 Tropical completion and wall pairs

Let Γ be the additive group of continuous functions $f : [0, \infty) \rightarrow \mathbb{R}$ that are piecewise affine with finitely many rational breakpoints, rational slopes and intercepts, and are eventually constant.

Pointwise maximum, minimum, negation, and division by two preserve Γ . Define

$$\mathbb{P}_{\max} = \Gamma \sqcup \{-\infty\}, \quad f \oplus g = \max(f, g), \quad f \odot g = f + g.$$

This is an entire zerosumfree idempotent semifield. Its finite elements are all units and its square map is surjective. Negation gives the dual min-plus semifield $\mathbb{P}_{\min} = \Gamma \sqcup \{+\infty\}$.

Let $L, R \in \Gamma$. On the free rank-one $\mathbb{P}_{\max}$ -module define

$$q_L(x) = x^{\odot 2} \odot L, \quad b_R(x, y) = x \odot y \odot R.$$

Theorem 5.1 (Wall pair and gap classification). *The pair (q_L, b_R) is quadratic exactly when $R \leq L$ pointwise. It is regular, and*

$$Q(L, R) \cong Q(L', R') \iff L - R = L' - R'.$$

Proof. Homogeneity and bilinearity are immediate. Pointwise,

$$\begin{aligned} q_L(x \oplus y) &= \max(2x + L, 2y + L), \\ q_L(x) \oplus q_L(y) \oplus b_R(x, y) &= \max(2x + L, 2y + L, x + y + R). \end{aligned}$$

If $R \leq L$, the third term is at most the maximum of the first two. Taking $x = y$ equal to the tropical multiplicative unit proves the converse.

The Gram coefficient R is finite and hence a unit, proving regularity. Every rank-one automorphism is multiplication by a finite unit u , which is an isometry precisely when

$$L = L' + 2u, \quad R = R' + 2u.$$

This preserves the gap. Conversely, equal gaps give such a unit with $u = (L - L')/2$. $\square$

The left and right thermograph walls of a short game satisfy $R_G \leq L_G$ and belong to Γ [8]. Thus they define $Q(G) = Q(L_G, R_G)$, whose isometry class is exactly the gap $D_G = L_G - R_G$. The dual min-plus construction records the same gap after negation.

5.2 The disjunctive-sum obstruction

In the stable group of Definition 2.4, distinct rank-one loop types survive freely: a basic metabolic regular object has only edge blocks by Theorem 2.5. The necessary basepoint reduction is

$$\Omega(G) = [Q(G)] - [Q(0)].$$

Theorem 5.2 (Hot-switch obstruction). *The map Ω is not additive under disjunctive sum. Consequently the paired max/min wall class is not an additive stable game invariant.*

Proof. Take the switch $X = \{1 \mid -1\}$. Its walls and gap are

$$L_X(t) = \max(1 - t, 0), \quad R_X(t) = -\max(1 - t, 0), \quad D_X(t) = 2\max(1 - t, 0).$$

The zero game has gap zero. Theorem 5.1 and freeness of the rank-one generators imply that $\Omega(X) \neq 0$ and has infinite order.

But $-X = X$, so $X + X = 0$ in the Conway game group [4]. Additivity would give $2\Omega(X) = \Omega(0) = 0$, a contradiction. Projection of a paired max/min invariant to its max component would be additive, so the same contradiction rules out the pair. A synchronized quotient can evade it only by killing the nonzero gap atom exhibited above. $\square$

There is a complementary freezing obstruction at temperature zero. Any additive invariant into a cancellative target that depends only on the frozen thermograph assigns the same value to $\uparrow$ and $2\uparrow$. The identity $x = 2x$ then cancels to $x = 0$. Thus wall-only additivity kills the entire common infinitesimal wall class even before the hot-switch argument.

6 Formal verification and implementation scope

The import-only entry point `Ogdoad.Papers.SemiringStability` exposes the Lean module `Ogdoad.SemiringQuadratic` [1], which checks:

1. uniqueness and diagonal balance of companions over additively cancellative semirings;
2. that finite basis labels and Gram coefficients determine the entire quadratic pair;
3. the no-half-of-one, no-nonzero-self-double, and trivial-unit properties of the polynomial model

$$\text{MvPolynomial}(\text{Fin } d, \mathbb{N})$$

for $\mathcal{H}_d$;

4. that every two-sided invertible square matrix over that polynomial semiring is a permutation matrix;
5. the end-to-end polynomial proxy theorem

`hessenbergPolynomial_regular_isometric_hyperbolic,`

which shows that bijectivity of the companion adjoint implies even rank and produces an explicit coefficient calculation together with an actual linear isometry of quadratic pairs to a canonical orthogonal sum of hyperbolic planes;

6. that an additive invariant into a cancellative monoid vanishes on any observation identified with its double; and
7. Mathlib’s classical diagonalization under $2 \neq 0$, together with an explicit additive equivalence between the exact cyclic/hyperbolic/parity presentation of the ring-image group and $\mathbb{Z}/2$.

Thus the kernel now checks the regular hyperbolic classification over the concrete polynomial model from the same input notion of regularity used in the paper; it no longer assumes the permutation-Gram, matching, or coefficient-to-isometry joins. The Cantor-normal-form identification with the actual ordinal fragment, the connected-graph stable quotient and Witt-group packaging, the construction of the concrete supertropical semiring and its two scalar-extension presentations, and standard thermograph facts remain proved here or cited external mathematics, not Lean theorems. For Theorem 4.4, Lean checks classical diagonalization and the exact final group equivalence, while the supertropical normalization and quotient-presentation fields are the explicit remaining bridge.

Ogdoad’s current `Ordinal` backend implements Conway nim arithmetic and ordinary noncommutative Cantor arithmetic. It is not a Hessenberg semiring. Likewise, the current finite-valued piecewise-linear wall type has no global tropical zero. Reusing either type silently would violate the coefficient hypotheses above. A future implementation should add a distinct finite-CNF natural-arithmetic type and the explicit one-point completions $\mathbb{P}_{\max}, \mathbb{P}_{\min}$.

7 Conclusion

Unique bases turn the absence of subtraction from an ambiguity into a classification tool: stable quadratic pairs become connected weighted graphs modulo a scalar gauge. In the Hessenberg fragments this yields both extremes at once. The regular Witt theory is zero, while the singular stable

envelope is a large free invariant with a finite decision procedure. Over the rational supertropical cover the same theorem gives exact blocks, companion ambiguity, and metabolic relations. Coefficientwise supervaluation remembers a chosen classical basis, but after imposing all source basis changes the universal classical image is exactly rank parity in characteristic different from two.

Thermograph walls complete the picture. After the minimal tropical completion they form honest regular rank-one pairs, and their isometry class is exactly the wall gap. Yet the self-negative hot switch makes that nonzero class incompatible with additivity. The common lesson is precise: the graph classification survives without subtraction, but classical basis invariance and game addition impose further quotient relations that the intrinsic semiring theory neither supplies nor conceals.

References

- [1] a9lim. Ogload: Clifford algebras, quadratic forms, and combinatorial games. Software and formalization, 2026.
- [2] Cahit Arf. Untersuchungen über quadratische formen in körpern der charakteristik 2, teil i. *Journal für die reine und angewandte Mathematik*, 183:148–167, 1941.
- [3] Philip W. Carruth. Arithmetic of ordinals with applications to the theory of ordered abelian groups. *Bulletin of the American Mathematical Society*, 48(4):262–271, 1942.
- [4] John H. Conway. *On Numbers and Games*. Academic Press, London, 1976.
- [5] Zur Izhakian, Manfred Knebusch, and Louis Rowen. Quadratic and symmetric bilinear forms on modules with unique base over a semiring. *Documenta Mathematica*, 21:773–808, 2016.
- [6] Zur Izhakian, Manfred Knebusch, and Louis Rowen. Supertropical quadratic forms I. *Journal of Pure and Applied Algebra*, 220(1):61–93, 2016.
- [7] Zur Izhakian, Manfred Knebusch, and Louis Rowen. Supertropical quadratic forms II: Tropical trigonometry and applications. *International Journal of Algebra and Computation*, 28(8):1633–1676, 2018.
- [8] Aaron N. Siegel. *Combinatorial Game Theory*, volume 146 of *Graduate Studies in Mathematics*. American Mathematical Society, 2013.